

Stability Regions for Standing Balance of Biped Humanoid Robots

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Abstract—Based on the liner inverted pendulum model (LIPM) for the dynamics of biped humanoid robots, an analytical method for computing stability regions relevant to standing balance of the biped humanoid robots is introduced in this paper. More precisely, two types of the stability regions are discussed in this paper with the consideration of that the zero moment point (ZMP) should be located in the supporting region to guarantee stable standing of the biped humanoid robots. First, assuming no external disturbances affecting the motion of the biped humanoid robots, the set of the initial values of the center of mass (CoM) position and velocity with which the location of the ZMP is limited to be inside the supporting region can be explicitly obtained by solving a finite number of linear inequalities. Second, two admissible sets of external force disturbances (impulse and finite energy) with which the ZMP does not deviate from the supporting region are characterized by solving finite number of linear inequalities or the discrete-time Lyapunov equation, respectively. The validity and effectiveness of the analytical method proposed in this paper are verified through a simulation result.

I. INTRODUCTION

The study on biped humanoid robots with their balance problem has been studied deeply for many years; e.g., feedback linearization method was introduced in [1] while computation method for forces of constraints was derived in [2]. These studies described the motion and balance of the biped humanoid robots through a simple model such as planar inverted pendulum.

On the other hand, the concept of the zero moment point (ZMP) defined as the center point of the reaction forces, was used to stability analysis of the biped humanoid robots in [3] for the first time. Because the ZMP is directly related to their balance and stability, a number of studies associated with the treatment of the ZMP have been subsequently presented. For example, the ZMP of the biped humanoid robots was regarded in [4]–[6] as the state variable to be controlled, while it was considered in [7],[8] as the control input by which their center of mass (CoM) is to be adequately regulated. Furthermore, in [9]–[12], the ZMP location is controlled to be inside the supporting region of the biped humanoid robots for their stable balance.

Based on the argument associated with the ZMP location for stable balance of the biped humanoid robots, the methods for computing stability regions are derived in [12]–[15]. More precisely, the regions of the real-time CoM position and velocity by which the associated torque input is saturated

with the center of pressure (equivalently to the ZMP) at the edge of the foot are obtained in [12] based on ankle or hip strategy. The set of the initial values of the CoM position and velocity with which the ZMP should be located inside the supporting region is derived in [13]–[15] by using the concept of the maximal output admissible (MOA) set [16] on the liner inverted pendulum model (LIPM) [17] of the biped humanoid robots. However, the method for obtaining the former set [12] carries out somewhat conservative computations through the boundary conditions of the supporting region. Furthermore, the concept of the MOA set used in the latter set [13]–[15] always involves the number of iterations, but no argument about the detailed value of the iterations by which the computed MOA set is guaranteed to coincide with the exact MOA set is provided in [13]–[15].

In connection with this, this paper provides an analytical method for computing the stability regions for stable standing balance of the biped humanoid robots based on their LIPM representation. The CoM position and velocity are treated as the state variable while the ZMP is regarded as the control input, as in [7],[8],[13]–[15]. This treatment allows us to describe the supporting region as a constraint on the control input. In this regard, two types of the stability regions are dealt with in this paper. First, assuming no external force disturbance affecting the motion of the biped humanoid robots, the set of the initial state values with which the constraint on the control input is always satisfied is considered in this paper. Second, two admissible sets of external force disturbances those of impulse and finite energy with which the constraint is always satisfied are characterized. This paper reduces the computation problems of these stability regions to the l_∞ norm analysis problem, which is also known as the l_1 problem [18]–[28] in the control theory fields. More precisely, the former set is explicitly obtained by solving finite number of linear inequalities while the latter sets are characterized by solving finite number of linear inequalities and the discrete-time Lyapunov equation, respectively.

The organization of this paper is as follows. In Section II, we first review the behavior of the biped humanoid robots through the relation of the CoM and the ZMP. We then formulate the problem definition for stable standing balance of the biped humanoid robots in Section III. We next provide in Section IV the main results of this paper, i.e., the computation methods for the stability regions. Section V gives some simulation results to demonstrate the validity and effectiveness of the computation methods. Finally, we state concluding remarks in Section VI.

In the following, the notations $\mathbb{N}$, $\mathbb{R}^\nu$ and $\delta(t)$ are used to denote the set of positive integers, the set of ν -dimensional

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real numbers and the impulse function occurring at $t = 0$, respectively, while the notation $\mathbb{N}_0$ is further used to imply $\mathbb{N} \cup \{0\}$. The ∞ -norm and 2-norm of a finite-dimensional vector are denoted by $|\cdot|_\infty$ and $|\cdot|_2$, respectively. i.e.,

$$|f|_\infty := \max_i |f_i|, \quad |f|_2 := (f^T f)^{1/2}$$

The notation $\|\cdot\|_\infty$ is used to mean either the l_∞ norm of a vector-valued sequence, i.e.,

$$\|f(\cdot)\|_\infty := \sup_{k \in \mathbb{N}_0} |f(k)|_\infty,$$

or the ∞ -norm of a finite-dimensional matrix, whose distinction will be clear from the context. The l_2 norm of a vector-valued sequence is denoted by $\|\cdot\|_2$, i.e.,

$$\|f(\cdot)\|_2 := \left(\sum_{k=0}^{\infty} |f(k)|_2^2 \right)^{1/2}$$

II. CoM-ZMP DYNAMICS OF BIPED ROBOTS

Let us consider a biped humanoid robot standing on a flat ground as shown in Fig. 1, where the x - y plane, S , and z -axis are set to denote the flat ground, supporting region, and vertical direction, respectively. The balance/stability issues of the biped humanoid robot have been dealt with deeply for many years and its CoM-ZMP behavior described by the LIPM has been widely used for simplifying the associated control and planning problems.

A. Description of Horizontal Motion via LIPM

This subsection describes the horizontal motion of biped humanoid robots by using the well-known LIPM, which describes the CoM dynamics under the assumption of a constant height of the CoM and zero angular momentum. In this paper, the slightly modified version of the LIPM used in [11] is considered to include an external force disturbance such as a push, which is described by

$$G_c : \begin{cases} m\ddot{c}_x = \frac{mg}{c_{z0}}(c_x - p_x) + f_x \\ m\ddot{c}_y = \frac{mg}{c_{z0}}(c_y - p_y) + f_y \end{cases} \quad (1)$$

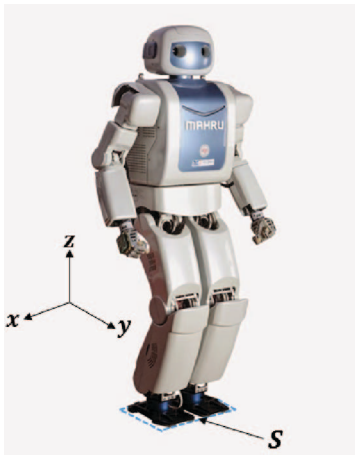


Fig. 1. Biped humanoid robot standing on flat ground.

where $c := [c_x \ c_y]^T \in \mathbb{R}^2$ is the horizontal position of the CoM at a constant height c_{z0} , $p := [p_x \ p_y]^T \in \mathbb{R}^2$ is the position of the ZMP, g is the gravitational acceleration and $[f_x \ f_y]^T \in \mathbb{R}^2$ is the external force applied at the CoM of the robot. Here, the first equation of (1) can be further represented by

$$\dot{x} = Ax + B_1 w + B_2 u \quad (2)$$

where $x := [c_x \ \dot{c}_x]^T \in \mathbb{R}^2$ is the state variable, $w := f_x \in \mathbb{R}$ is the disturbance, $u := p_x \in \mathbb{R}$ is the control input and

$$A = \begin{bmatrix} 0 & 1 \\ \omega_0^2 & 0 \end{bmatrix}, \quad B_1 = \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix}, \quad B_2 = \begin{bmatrix} 0 \\ -\omega_0^2 \end{bmatrix} \quad (3)$$

with $\omega_0^2 := g/c_{z0}$. The linear time-invariant (LTI) nature of the LIPM (2) allows us to develop analytical and theoretical insights into stability analysis of the biped humanoid robots. For the second equation of (1), we can obtain the same form by defining $x := [c_y \ \dot{c}_y]^T$, $u := p_y$, and $w := f_y$.

B. Stability Conditions for Standing Balance of Biped Humanoid Robots

Let us set the coordinates of the x - y plane together with the supporting region S as shown in Fig. 2; the center of S , whose shape is comprised of rectangle with width and height lengths of α_x and α_y , respectively, is set to the origin $(0, 0)$. Here, the following two conditions for stable standing balance of the biped humanoid robots are concerned with in this paper:

- (i) The position and velocity of the CoM tend to zero for its arbitrary bounded initial value when there is no external force disturbance.
- (ii) The ZMP is always located inside the supporting region S .

These conditions are also mathematically represented by

- (i)* $x(t) \rightarrow 0$ ($t \rightarrow \infty$) for an arbitrary $x(0) \in \mathbb{R}^2$ such as $|x(0)|_\infty < \infty$ when $w(t) \equiv 0$ ($\forall t \geq 0$).
- (ii)* $u(t) \in S$ ($\forall t \geq 0$).

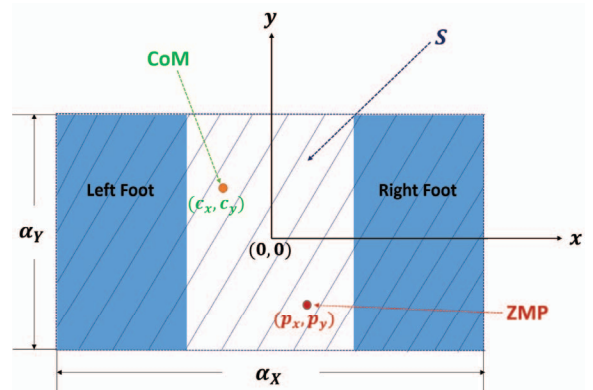


Fig. 2. Coordinates of x - y plane and supporting region S .

III. PROBLEM DEFINITION FOR STANDING BALANCE

As a preliminary step to formulate the problem definition for stable standing balance of the biped humanoid robots by using the state-space equation (2), we first give the notion of the pathological sampling frequency as follows.

Definition 1 ([29]): Let us consider the continuous-time system

$$\dot{x}_c = A_c x_c + B_c u_c \quad (4)$$

Then, the sampling frequency $\omega_s (= 2\pi/T)$ is pathological (relative to A_c) if A_c has two eigenvalues with equal real parts and imaginary parts that differ by an integral multiple of ω_s . Otherwise, the sampling frequency is non-pathological.

We then assume the non-pathological sampling frequency in Definition 1 throughout the paper. With the non-pathological sampling period T (i.e., $2\pi/T$ is non-pathological sampling frequency) relative to A , we have the discretized model of the continuous-time LIPM (2) as

$$\xi(k+1) = A_d \xi(k) + B_{1d} \eta(k) + B_{2d} \psi(k) \quad (5)$$

where $A_d := \exp(AT)$ and

$$B_{1d} := \int_0^T \exp(A\theta) B_1 d\theta, \quad B_{2d} := \int_0^T \exp(A\theta) B_2 d\theta \quad (6)$$

with $\xi_k := x(kT)$, $\eta_k := w(kT)$ and $\psi_k := u(kT)$.

We reformulate the stability conditions (i)* and (ii)* for the discrete-time state-space representation (5) as follows:

- (I) $\xi(k) \rightarrow 0$ ($k \rightarrow \infty$) for an arbitrary $\xi(0) \in \mathbb{R}^2$ such as $\|\xi(0)\|_\infty < \infty$ when $\eta(k) \equiv 0$ ($\forall k \in \mathbb{N}_0$).
- (II) $\psi(k) \in S$ ($\forall k \in \mathbb{N}_0$).

If we design the full-state feedback controller such as $\psi(k) = K\xi(k)$, then the closed-loop behavior of (5) is described by

$$\begin{aligned} \xi(k+1) &= (A_d + B_{2d}K)\xi(k) + B_{1d}\eta(k) \\ &=: A_{2d}\xi(k) + B_{1d}\eta(k) \end{aligned} \quad (7)$$

Here, we can make A_{2d} have all its eigenvalues in the open unit disc since the pair (A_d, B_{2d}) is controllable because of the assumption of the non-pathological sampling period. This clearly means that the condition (I) is satisfied by using the full-state feedback controller K adequately. From the controllability property of the pair (A_d, B_{2d}) , we further assume that A_{2d} is a diagonalizable matrix throughout the paper. To discuss the condition (II) on (7), we next provide the discrete-time LTI system

$$G_d : \begin{cases} \xi(k+1) = A_{2d}\xi(k) + B_{1d}\eta(k) \\ \zeta(k) = C_d\xi(k) \end{cases} \quad (8)$$

where $C_d := \frac{2}{\alpha_x} K$. Then, it readily follows that

The condition (II) is satisfied $\Leftrightarrow \|\zeta\|_\infty \leq 1$.

IV. COMPUTATION OF STABILITY REGIONS

This section provides the main results of this paper, i.e., the computation methods for the stability regions in connection with the l_1 analysis problem of $\|\zeta\|_\infty \leq 1$. To this end, we first represent the behavior of $\zeta(k)$ by

$$\zeta(k) = C_d A_{2d}^k \xi(0) + \sum_{i=0}^{k-1} C_d A_{2d}^{k-i-1} B_{1d} \eta(i) \quad (9)$$

By noting (9), this section deals with two types of the stability regions with respect to $\xi(0)$ and $\zeta(k)$, respectively. More precisely, we first assume no external force disturbances, i.e., $\eta(k) \equiv 0$ ($\forall k \in \mathbb{N}_0$), and consider the set of the initial values $\xi(0) \in \mathbb{R}^2$ with which $\|\zeta\|_\infty \leq 1$ is satisfied. We next assume the stationary state at $k = 0$, i.e., $\xi(0) = 0$, and characterize two admissible sets of the disturbance forces η those of impulse and finite energy with which $\|\zeta\|_\infty \leq 1$ is guaranteed.

A. Set of Initial Values of CoM Position and Velocity

By substituting $\eta(k) \equiv 0$ ($\forall k \in \mathbb{N}_0$) into (9), we obtain

$$\zeta(k) = C_d A_{2d}^k \xi(0) \quad (10)$$

and consider the set of $\xi(0)$ by which the l_∞ norm of ζ in (10) is not larger than 1. In connection with this, let us introduce for $N \in \mathbb{N}_0$ the following set:

$$\mathcal{O}_N = \{\xi(0) \mid |C_d A_{2d}^k \xi(0)| \leq 1 \text{ for } k = 0, \dots, N\} \quad (11)$$

The purpose of this subsection is to obtain $\mathcal{O}_\infty$ explicitly, but it is a non-trivial task since a direct treatment of $\mathcal{O}_\infty$ involves an infinite number of linear inequalities. In this sense, the pioneering studies [13]–[15] associated with the computation of $\mathcal{O}_\infty$ argue that $\mathcal{O}_\infty$ can be computed by finite number of iterations, but any argument about the detailed value of the iterations is not provided in those studies.

With this in mind, this subsection provides an explicit method for computing $\mathcal{O}_\infty$ with a finite number of linear inequalities. To this end, we can first easily compute $\mathcal{O}_0$ by solving 1 linear inequalities with 2 variables (because $C_d \in \mathbb{R}^{1 \times 2}$ and $\xi(0) \in \mathbb{R}^2$) and define β as

$$\beta := \max_{\xi(0) \in \mathcal{O}_0} \|\xi(0)\|_\infty \quad (12)$$

We next diagonalize A_{2d} by

$$A_{2d} = P \text{diag}[\lambda_1, \lambda_2] P^{-1} \quad (13)$$

where λ_1 and λ_2 are eigenvalues of A_{2d} , and let ρ be the spectrum radius of A_{2d} , i.e., $\rho = \max\{|\lambda_1|, |\lambda_2|\}$. Then, we have the following result.

Theorem 1: Let us define $N_0 \in \mathbb{N}_0$ as

$$N_0 := \max \left\{ 0, \left\lceil \log_\rho \frac{1}{\|C_d P\|_\infty \cdot \|P^{-1}\|_\infty \cdot \beta} \right\rceil \right\} \quad (14)$$

where $\lceil \cdot \rceil$ denotes the smallest integer which is not smaller than $(\cdot)$. We then obtain

$$\mathcal{O}_{N_0} = \mathcal{O}_\infty \quad (15)$$

We can see from Theorem 1 that $\mathcal{O}_\infty$ can be explicitly computed by solving N_0 linear inequalities with 2 variables.

B. Admissible External Force Disturbances

In this subsection, we assume the stationary state at $k = 0$, i.e., $\xi(0) = 0$, and consider the simplified version of (9) as

$$\zeta(k) = \sum_{i=0}^{k-1} C_d A_{2d}^{k-i-1} B_{1d} \eta(i) \quad (16)$$

This subsection aims at characterizing $\eta(k)$ ($k = 0, 1, \dots$) with which the l_∞ norm of ζ in (16) does not exceed 1. More precisely, two types of η are dealt with in this paper:

- ① impulsive disturbance, i.e., $\eta(0) \neq 0$ and $\eta(k) = 0$ for all $k \in \mathbb{N}$.
- ② disturbance of finite energy, i.e., $\|\eta\|_2 < \infty$.

The stability region for the first type of the disturbance corresponds to stability performance against an impulsive force while that for the second type of the disturbance could be used for the performance criterion for a disturbance force with a finite energy.

1) *Characterization of Impulse Disturbance:* With the assumption of that $\eta(0) \neq 0$ and $\eta(k) = 0$ for all $k \in \mathbb{N}$, we characterize $\eta(0)$ by which the l_∞ norm $\|\zeta\|_\infty$ in (16) does not exceed 1. More precisely, we define the set $\mathcal{O}_N^\eta$ as

$$\mathcal{O}_N^\eta = \{\eta(0) \mid |C_d A_{2d}^k B_{1d} \eta(0)| \leq 1 \text{ for } k = 0, \dots, N\} \quad (17)$$

and aim at computing $\mathcal{O}_\infty^\eta$ exactly. Here, we could see from the comparison between (10) and (17) that the characterization procedure is essentially the same as the procedure of computing $\mathcal{O}_\infty (= \mathcal{O}_{N_0})$ discussed in the previous subsection. In connection with this, we are led to the following result.

Theorem 2: Define $N_\eta \in \mathbb{N}_0$ as

$$N_\eta := \max \left\{ 0, \left\lceil \log_\rho \frac{1}{\|C_d P\|_\infty \cdot \|P^{-1} B_{1d}\|_\infty \cdot \beta_\eta} \right\rceil \right\} \quad (18)$$

where

$$\beta_\eta := \max_{\eta(0) \in \mathcal{O}_0^\eta} |\eta(0)| \quad (19)$$

Then, we have

$$\mathcal{O}_{N_\eta}^\eta = \mathcal{O}_\infty^\eta \quad (20)$$

This theorem characterizes the impulse disturbance with which the stability criteria $\|\zeta\|_\infty \leq 1$ is satisfied and clearly shows that $\mathcal{O}_\infty^\eta$ can be explicitly obtained through N_η linear inequalities.

2) *Characterization of Disturbance of Finite Energy:* This subsection considers the worst case of η associated with the marginal stability in which $\|\zeta\|_\infty$ can be equal to 1. More precisely, we are in a position to compute γ^* defined as

$$\gamma^* := \max \gamma \quad \text{such that} \quad \sup_{\|\eta\|_2 \leq \gamma} \|\zeta\|_\infty \leq 1 \quad (21)$$

Because the mapping from η to ζ is linear as shown in (16), γ^* can be computed as the reciprocal of the induced norm from l_2 to l_∞ , i.e.,

$$\|G_d\|_{\infty/2} := \sup_{\|\eta\|_2 \leq 1} \|\zeta\|_\infty \quad (22)$$

and $1/\gamma^* = \|G_d\|_{\infty/2}$. Throughout the paper, we call $\|G_d\|_{\infty/2}$ the l_∞/l_2 -induced norm of G_d . The computation procedure of $\|G_d\|_{\infty/2}$ introduced in this paper can be interpreted as a slightly modified version of the similar argument in [30] that deals with only single-input/single-output (SISO) case. The l_∞/l_2 -induced norm of G_d can be described by

$$\begin{aligned} \|G_d\|_{\infty/2} &= \sup_{\|\eta\|_2 \leq 1} \|\zeta\|_\infty \\ &= \left(\sum_{i=0}^{\infty} C_d A_{2d}^i B_{1d} B_{1d}^T (A_{2d}^i)^T C_d^T \right)^{1/2} \end{aligned} \quad (23)$$

Here, the summation of infinite series in (23) is given by

$$\left(\sum_{i=0}^{\infty} C_d A_{2d}^i B_{1d} B_{1d}^T (A_{2d}^i)^T C_d^T \right)^{1/2} = (C_d X_d C_d^T)^{1/2} \quad (24)$$

where X_d is the solution of the discrete-time Lyapunov equation

$$A_{2d} X_d A_{2d}^T - X_d + B_{1d} B_{1d}^T = 0 \quad (25)$$

We then have from (23) and (24) the following result.

Theorem 3: The l_∞/l_2 -induced norm $\|G_d\|_{\infty/2}$ coincides with

$$\|G_d\|_{\infty/2} = (C_d X_d C_d^T)^{1/2} \quad (26)$$

and γ^* is computed by

$$\gamma^* = 1/(C_d X_d C_d^T)^{1/2} \quad (27)$$

V. SIMULATION RESULTS

This section aims at demonstrating the effectiveness and validity of the computation methods developed in the previous section through some simulation results, but this section is confined itself to the argument associated with Theorem 1 for a limited space. We examine standing balance of the KIST humanoid robot 'Mahru' as shown in Fig. 1, where the detailed specifications are given in Table I.

We obtain the discrete-time system G_d of (5) with the non-pathological sampling period T by which the pair (A_d, B_{2d}) is to be controllable. We then adequately design the full-state feedback controller K such that $A_{2d} = A_d + B_{2d} K$ has all its eigenvalues in the open unit disc. With A_{2d}

TABLE I
SPECIFICATIONS OF MAHRU-R.

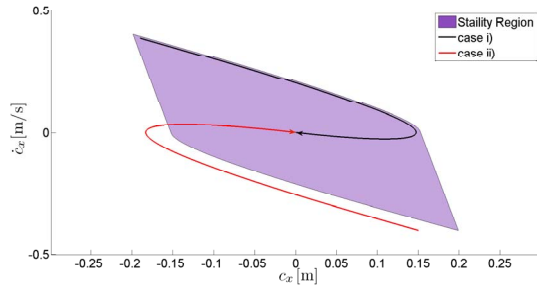
Height	1.35 m
CoM Height	0.697 m
Weight	50 kg
Supporting Region	0.31 m × 0.23 m

determined as mentioned above, we explicitly obtain the stability region $\mathcal{O}_\infty$ based on the arguments in Theorems 1. Indeed, the validity of the stability regions together with the stability criterion is demonstrated by giving the associated ZMP trajectories to confirm whether or not the ZMP is always stayed inside the supporting region.

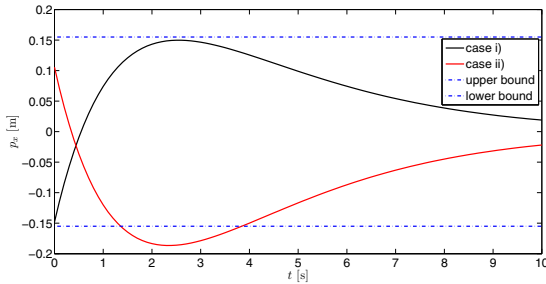
By using the arguments in Theorem 1, the stability region associated with the initial values of the CoM position and velocity are explicitly obtained as shown in Fig. 3 (a) and Fig. 4 (a) for $(c_x, \dot{c}_x)$ and $(c_y, \dot{c}_y)$, respectively. In these figures, the simulation results for the CoM trajectories are also given for two cases to examine the validity of the obtained stability region; i) the CoM position and velocity start inside the stability region, or ii) the CoM initial value start outside the stability region. The ZMP trajectories associated with these two cases are also given in Fig. 3 (b) and Fig. 4 (b) for p_x and p_y , respectively.

It can be observed from Fig. 3 (a) and Fig. 4 (a) for the case i) that the CoM trajectories always stay inside their stability regions if they start in those regions. Furthermore, not surprisingly, we can also see from these figures for the case ii) that the CoM trajectories converge to the origin $(0, 0)$ even if they start outside of the stability region; this is because A_{2d} is a stable matrix.

We can also see from Fig. 3 (b) and Fig. 4 (b) that the ZMP trajectories are always located inside the supporting region if the associated CoM initial values are inside the stability regions. More interestingly, even though the ZMP

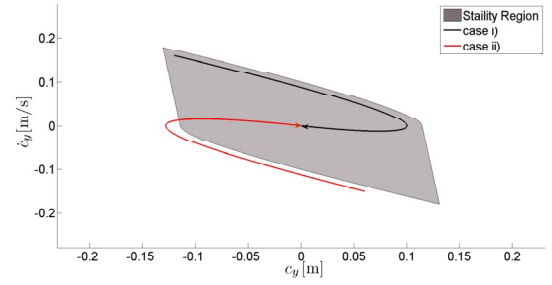


(a) Stability region for initial values of $(c_x, \dot{c}_x)$. i) The CoM trajectory $(c_x, \dot{c}_x)$ always stays inside the stability region. ii) The CoM trajectory $(c_x, \dot{c}_x)$ starts outside the stability region.

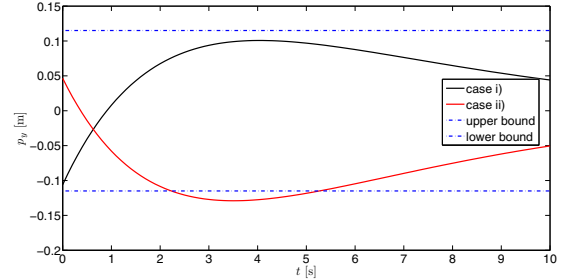


(b) ZMP trajectories for two cases. i) p_x with the initial value of $(c_x, \dot{c}_x)$ inside the stability region. ii) p_x with the initial value of $(c_x, \dot{c}_x)$ outside the stability region.

Fig. 3. Stability region for the CoM initial values and the ZMP trajectories in x -domain.



(a) Stability region for initial values of $(c_y, \dot{c}_y)$. i) The CoM trajectory $(c_y, \dot{c}_y)$ always stays inside the stability region. ii) The CoM trajectory $(c_y, \dot{c}_y)$ starts outside the stability region.



(b) ZMP trajectories for two cases. i) p_y with the initial value of $(c_y, \dot{c}_y)$ inside the stability region. ii) p_y with the initial value of $(c_y, \dot{c}_y)$ outside the stability region.

Fig. 4. Stability region for the CoM initial values and the ZMP trajectories in y -domain.

starting points for the case i) are closer to the boundaries of the supporting region than those for the case ii), the ZMP trajectories for the former are always limited under the supporting region while those for the latter do not satisfy such a limitation.

These observations clearly demonstrate the effectiveness and validity of the computation method discussed in Theorem 1 that corresponds to the stability region with respect to the initial values of the CoM position and velocity.

VI. CONCLUDING REMARKS

This paper dealt with stability regions for standing balance of biped humanoid robots. Based on the linear inverted pendulum model (LIPM) for the dynamics of the biped humanoid robots, two types of the stability region were discussed in this paper. The key idea for dealing with the stability regions is that the zero moment point (ZMP) should be located inside the supporting region. With this in mind, assuming no external disturbances, we first explicitly computed the set of the initial values of the center of mass (CoM) position and velocity by solving finite number of linear inequalities. We next characterized two admissible sets of external disturbance forces those of impulse and finite energy by solving finite number of linear inequalities and the discrete-time Lyapunov equation, respectively. In this sense, the contributions of the present paper can be interpreted as follows: i) giving the explicit computation method of the CoM initial values through finite number

of linear inequalities while no detailed argument about the associated iteration number for the computation is given in the pioneering studies [13]–[15], ii) characterizing the admissible external force disturbances for the first time. The success in this paper in the treatment of the stability regions is believed to contribute to wider applications of the biped humanoid robots to practical problems. Finally, we expect that the CoM initial values and the admissible disturbance forces could be treated at the same time by following the same line of the arguments in this paper, but developing such a method might lead to involved arguments, and thus this interesting topic is left for future studies.

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